PELAYO PACHE RODRÍGUEZ

INDUSTRIAL ELECTRONICS AND AUTOMATION ENGINEERING STUDENT

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Education

UNIVERSITY OF OVIEDO (EPI Gijón) **Gijón, Spain**

B.Sc. in Industrial Electronics and Automation Engineering Sept. 2022 – Present (Expected Jun. 2027)

- **Academic Record:** Cumulative GPA: **7.49 / 10.0** (Equivalent to UK 2:1 Honours / ∼3.2 US GPA).
- **Bachelor's Thesis (TFG):** Conducting TFG in direct collaboration with the **University of León** in the field of **Aeronautical Engineering**.
- **Relevant Coursework: Digital Electronics**, **Embedded Systems Microelectronics**, **Power Electronics**, **Control Engineering**, Analog Circuit Design, Electronic Instrumentation, and Industrial Automation.

LIVERPOOL JOHN MOORES UNIVERSITY (LJMU) **Liverpool, United Kingdom**

Academic Exchange Year – Faculty of Engineering and Technology *Sept. 2025 – Sept.2026*

- **Academic Classification: Distinction / First Class Honours** grade (Equivalent Spanish average grade: **9.15 / 10.0**).
- Full academic year taught entirely in English, focused on advanced embedded automation, sensor integration, applied control theory, and development in international industrial environments.

COLEGIO SANTO DOMINGO DE GUZMÁN (Dominicos) **Oviedo, Spain**

High School Diploma in Science and Technology *Sept. 2020 – May 2022*

Technical Experience & Engineering Projects

RESEARCH COLLABORATION GRANT **Oviedo / Gijón, Spain**

Selected for Top Research Collaboration Project *Sept. 2026 – Present*

- **Innovation Selection:** Development of laboratory test equipment for satellite constellation release testing or solar panel deployment in simulated microgravity.
- **Project Defense:** Delivery of a comprehensive project report for formal defense, endorsed by the faculty advisor for submission to the Ministry of Education.

ART – ASTURIAS RURAL TECH (UNIOVI EMPRENDE) **Oviedo / Gijón, Spain**

Co-Founder & Lead Technology Developer (Finalist Project – Top 12 Innovation) *Sept. 2026 – Present*

- **Top Innovation Finalist Selection:** Selected as **Finalists** among the **Top 12 Innovation Projects** in the official entrepreneurship program *Uniovi Emprende*, organized in strategic partnership with **Banco Santander** and leading regional business entities; qualified for and defended the initiative before an executive evaluation tribunal on **November 5, 2026**.
- **Direct Booking Platform & Digital Architecture:** Co-designed and deployed the web platform architecture and proprietary direct booking engine (ART Portal) engineered to eliminate abusive 15%–25% commissions from multinational OTAs (Booking, Airbnb), enabling rural accommodations and commerce to retain 100% of their net margin.
- **Field Methodology («Tú a Tú») & Commercial Pilot Validation:** Spearheaded on-the-ground technical consulting and local SEO diagnostics in rural municipalities; validated the model through an operational pilot with a traditional enterprise in Pola de Laviana, achieving a **4x increase in direct customer contacts and calls** via Google Maps optimization.

- **Strategic Financing & Market Modeling:** Structured a 4-tier scalable monetization model addressing a **EUR 22M TAM** (2,500+ rural businesses in Asturias), lowering client adoption barriers to **EUR 0 upfront cost** by channeling official *Kit Digital* subsidies (EUR 3,000–EUR 12,000 grants) to foster long-term rural business sustainability.

UNIOVI ETECH-RACING (FORMULA STUDENT) **Gijón, Spain**
Hardware & Low Voltage (LV) Electronics Engineer *Sept. 2024 – Mar. 2026*

- **PCB Design:** Schematics, layout routing, and validation of multi-layer Printe Circuit Boards (PCBs) using **Altium Designer** (Advanced Level) for the electric racecar's Low Voltage (LV architecture, ensuring signal integrity and compliance with Formula Student regulations.
- **Industrial Communications:** Implementation and debugging of the **CAN Bus** protocol (500 kbps) for real-time telemetry interconnection between the ECU, telemetry system, and BMS (*Battery Management System*).
- **Embedded Systems & Logic:** Microcontroller programming in **C/C++** and digital architecture design in **VHDL (Xilinx Vivado)** for high-speed sensor acquisition and actuator control.
- **Simulation & Electronic Modeling:** Simulation of electrical transients and power electronics in **PLECS**, **PSIM**, and **MATLAB & Simulink** to optimize energy efficiency.
- **Harnessing & Testing:** Fabrication of motorsport-grade wiring harnesses with EMI shielding. Bench testing and **soldering** (THT and SMD).

UNIOVI INNOVATION SKILLS **Gijón, Spain**
Technical Lead & Hardware Prototyping (1st Prize for Most Innovative Project)Academic Year 2023 – 2024

- Co-designed an industrial monitoring and predictive maintenance system in partnership with *Gijón Impulsa* and *Tresa Ascensores*.
- Developed the acquisition electronics and defended the prototype before an industrial evaluation committee, winning **1st Prize for Most Innovative Project**.

Additional Experience & Leadership

PRIVATE MATHEMATICS AND PHYSICS TUTOR **Oviedo, Spain**
Private Tutor *2022 – Present*

- Tailored one-on-one tutoring in calculus, linear algebra, and general physics to over 10 high school students, enhancing technical communication skills.

EDP (ELECTRICIDADE DE PORTUGAL) – FIDMA FAIR **Gijón, Spain**
FIDMA International Trade Fair Assistant *August 2023*

- Provided customer support and technical communication regarding renewable energy projects and innovation at the institutional EDP pavilion.

ALVAPAN **Oviedo, Spain**
Delivery Assistant (Summer Contract) *Summer 2022*

- Managed delivery routes, customer relations, and logistics under demanding working conditions.

Technical Skills & Competencies

- **Systems Engineering & MBSE: SysML v2** – Model-Based Systems Engineering (MBSE), with practical application experience in both specialized environments (**Dassault CATIA / MagicDraw**) and tool- agnostic modeling architectures.
- **Hardware Design & EDA: Altium Designer** (Advanced Level), KiCAD, Multi-layer PCBs, Schematic Capture, Electronic Prototyping, **Soldering** (THT and SMD), Bench Instrumentatio (Oscilloscopes, Logic Analyzers, Regulated Power Supplies).

- **Embedded Systems & Digital Logic:** Microcontrollers (**STM32**, **ARM Cortex**, **Microchip PIC** with **MPLAB**), **VHDL (Xilinx Vivado)**, **CAN Bus**, SPI I2C, UART, Digital Logic, Siemens PLCs (**TIA Portal**).
- **Simulation & Modeling: MATLAB & Simulink**, **PLECS**, **PSIM**,FUSION, INVENTOR, AutoCAD.
- **Programming Languages: C/C++**, **Python**, **Java**, **R**.
- **Soft Skills & Leadership:** Leadership, Teamwork, Technical Problem Solving, Troubleshooting, Adaptability & Fast Learning.
- **Languages:**

  Spanish: Native / Bilingual.

  English: C1 Full Professional Proficiency (Oxford ELLT C1; Cambridge B2; scheduled IELTS).

  French: Elementary / Basic Proficiency.

Certifications & Qualifications

- **British Airways – Engineering Job Simulation:** Aeronautical maintenance, systems reliability, and flight engineering (Forage).
- **Santander Open Academy Academic Excellence Scholarship:** Awarded for ERASMUS exchange students.
- **Santander Open Academy:** Artificial Intelligence with Gemini and Large Language Models.
- **Oxford ELLT (C1 Proficient):** University of Oxford / OI Digital Institute.
- **Cambridge First Certificate in English (FCE - B2):** University of Cambridge.
- **Driving License:** Full EU Class B.